Does aging microplastics alter how microplastics affect the physiology of a freshwater copepod

The Zooplankton Ecology & Adaptation Lab At UMass Lowell

Jackson Vanegas, Matthew Sasaki

UML Biology

Introduction

- Microplastics are increasing in abundance in aquatic systems
- Changes in food quality and quantity
- Copepods are important, highly selective aquatic grazers
- Biofilms can form on surface of microplastics
  - Nutritional benefit?
  - Increased exposure via reduced selectivity?

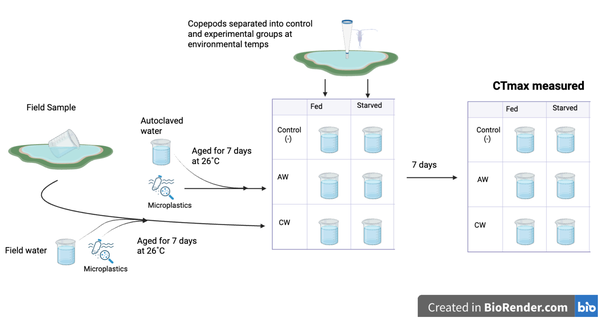
Central Question

Does microplastic aging alter how these particles affect the physiology of a common freshwater copepod?

Hypothesis

1. Exposure to microplastics will decrease the thermal limits in the freshwater copepod *Skistodiaptomus pallidus.*
2. The upper thermal limit for copepods exposed to aged microplastics will be higher than the upper thermal limit for those exposed to un-aged microplastics.
3. Because of the nutritional content of biofilms, starved copepods with aged microplastics will exhibit a smaller decrease in thermal limit relative to the control copepods, while starved copepods with un-aged microplastics will exhibit a sharp decrease in thermal limits.

Methods



Results

Figure 1: Descriptive figure caption here

Methods

- Provide methodological details here. Or better yet, use a schematic

Conclusion

- Conclusions here

Next Steps

Additional experimental replicates

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